

D459

Introduction to Systems Thinking and Applications

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A software startup founded by two former teachers finds that each satisfied user refers about two new users, who then refer others; monthly signups keep accelerating. The company's office is in a renovated warehouse. What type of loop drives this behavior?

- A. A balancing loop.
- B. A one-way cause-and-effect chain with no feedback.
- C. A reinforcing loop.
- D. A delay loop.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A thermostat turns the heat on when a room drops below 68°F and off when it rises above it. Which loop is this, and what behavior does it produce?

- A. A balancing loop that holds the temperature near the goal.
- B. A reinforcing loop that drives temperature ever higher.
- C. A balancing loop that produces exponential growth.
- D. No feedback at all; it is a simple switch.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A retailer with 40 stores orders more stock when shelves look empty, but deliveries take three weeks. Store managers, most with over ten years' experience, keep adding orders each week until the first shipment lands, then find themselves overstocked, cut orders sharply, and run short again. What best explains the cycle?

- A. Customer demand must be swinging in the same cycle.
- B. The delay between ordering and delivery causes overcorrection and oscillation.
- C. A reinforcing loop is driving runaway growth in stock.
- D. Managers lack experience with ordering.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A county animal shelter with a volunteer board holds 120 dogs on March 1. During March it takes in 85 dogs and places 110 through adoption and transfer; its new website launched mid-month. How many dogs are in the shelter at the end of March?

- A. 95
- B. 145
- C. 85
- D. 25

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

5. [TRANSFER — reversed stem / different variable]

A company's order backlog was 600 at the start of the week and 450 at the end. It received 900 new orders during the week. How many orders did it complete?

- A. 750
- B. 150
- C. 900
- D. 1,050

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

6.

A city water utility, which recently won a regional service award, faces rising demand. Options: raise the price per gallon by a few cents, add a new reporting dashboard, or change the utility's goal from "sell more water" to "deliver water services with the lowest total resource use." Which is the highest-leverage intervention?

- A. Adding the reporting dashboard.
- B. Raising the price per gallon.
- C. Building a larger reservoir, since physical capacity is always the strongest lever.
- D. Changing the utility's goal.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

7.

A factory manager with an MBA from a top school responds to missed deadlines by approving weekend overtime. Output rises for a few weeks, then fatigue drives up errors and rework, deadlines slip again, and more overtime is approved. Which archetype is this?

- A. A reinforcing growth loop, so more overtime will eventually solve it.
- B. Limits to growth.
- C. Fixes that fail.
- D. Shifting the burden.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

In the fixes-that-fail archetype, what makes the problem come back?

- A. People come to depend on the fix and the fundamental solution withers.
- B. The fix never works at all, even briefly.
- C. The fix runs into a resource limit that slows growth.
- D. An unintended side effect of the fix feeds back, after a delay, to worsen the original symptom.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

After a third data breach in two years at a mid-sized insurer with offices in four states, leaders learn that staff are rewarded for closing tickets fast, security reviews are optional, and many managers believe “security slows us down.” Using the iceberg model, at which level is “security slows us down”?

- A. Structures.
- B. Mental models.
- C. Patterns.
- D. Events.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which intervention works at the structure level of the iceberg?

- A.** Making security reviews mandatory and changing ticket-closing incentives.
- B.** Patching the server that was breached.
- C.** Tracking breach frequency on a dashboard.
- D.** Running a campaign to change beliefs about security.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A hospital with a new parking garage rewards each department for its own efficiency. The ER cuts wait times by moving patients to inpatient units faster, but inpatient units overflow and discharge delays grow; overall patient flow worsens. What principle explains this?

- A.** Every department improved its own metric, so the whole must have improved.
- B.** Optimizing parts separately can suboptimize the whole, because performance emerges from how parts interact.
- C.** The ER should cut wait times even further.
- D.** Inpatient units are simply understaffed; hire more nurses.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A meal-kit company's subscribers grew 15% a month for a year as marketing spend rose. Growth has now stalled even though marketing spending keeps climbing, and delivery complaints have tripled because the single packing center is at capacity. The company's app has a 4.7-star rating. What should leaders do?

- A.** Increase marketing spend again to restart growth.
- B.** Cut marketing to zero, since growth is over.
- C.** Relieve the packing-center constraint before pushing growth further.
- D.** Improve the app further, since customers rate it highly.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A software startup ~~founded by two former teachers~~ finds that each satisfied user refers about two new users, who then refer others; monthly signups keep accelerating. ~~The company's office is in a renovated warehouse.~~ What type of loop drives this behavior?

Target word: loop type **Decides:** More users produce more referrals, which produce more users: accelerating growth.

A	Inverted operation	Balancing loops pull toward a goal; nothing here is limiting growth yet.
B	Dropped element	Drops the link back from new users to more referrals.
C	Correct	Change feeds on itself and grows faster over time.
D	Adjacent category	Delays exist in many loops but are not a loop type; the behavior is reinforcing.

2. Answer A Concept 1 • transfer

A thermostat turns the heat on when a room drops below 68°F and off when it rises above it. Which loop is this, and what behavior does it produce?

Target word: reverse: from behavior to loop type **Decides:** The system acts to close the gap to a goal.

A	Correct	Goal-seeking correction is the signature of a balancing loop.
B	Inverted operation	The heater shuts off at the goal; nothing amplifies.
C	Confusion set	Right loop, wrong behavior; exponential growth belongs to reinforcing loops.
D	Dropped element	The room temperature feeds back to control the switch.

3. Answer B Concept 2

A retailer ~~with 40 stores~~ orders more stock when shelves look empty, but deliveries take three weeks. Store managers, ~~most with over ten years' experience~~, keep adding orders each week until the first shipment lands, then find themselves overstocked, cut orders sharply, and run short again. What best explains the cycle?

Target word: cause of the boom-bust cycle **Decides:** A three-week delay in the ordering loop.

A	Confusion set	Blames an outside event; the swings arise even with steady demand.
B	Correct	Managers act on a gap that orders already in transit will close.
C	Adjacent category	Stock swings up and down around a goal, which is a delayed balancing loop.
D	Camouflage	They are experienced; the structure, not skill, produces the swings.

4. Answer A Concept 3

A county animal shelter ~~with a volunteer board~~ holds 120 dogs on March 1. During March it takes in 85 dogs and places 110 through adoption and transfer; ~~its new website launched mid-month~~. How many dogs are in the shelter at the end of March?

Target word: ending stock **Decides:** Stock + inflow – outflow.

A	Correct	$120 + 85 - 110 = 95$.
B	Inverted operation	$120 + 110 - 85$ swaps inflow and outflow.
C	Wrong stem number	The month's inflow, not the ending stock.
D	Dropped element	The net flow ($110 - 85$) without the starting stock.

5. Answer D Concept 3 · transfer

A company's order backlog was 600 at the start of the week and 450 at the end. It received 900 new orders during the week. How many orders did it complete?

Target word: solve for the outflow **Decides:** Outflow = start + inflow – end.

A	Inverted operation	900 – 150 subtracts the backlog drop instead of adding it.
B	Dropped element	Only the change in the stock; ignores new orders.
C	Wrong stem number	The inflow, not the outflow.
D	Correct	$600 + 900 - 450 = 1,050$.

6. Answer D Concept 4

A city water utility, ~~which recently won a regional service award,~~ faces rising demand. Options: raise the price per gallon by a few cents, add a new reporting dashboard, or change the utility's goal from "sell more water" to "deliver water services with the lowest total resource use." Which is the highest-leverage intervention?

Target word: highest leverage point **Decides:** Changing the system's goal outranks parameters and information flows.

A	Confusion set	Information flows help but rank below changing the goal.
B	Adjacent category	A parameter change, which is low leverage.
C	Absolute language	Physical stocks are among the weakest leverage points.
D	Correct	Goals shape every rule and parameter below them.

7. Answer C Concept 5

A factory manager ~~with an MBA from a top school~~ responds to missed deadlines by approving weekend overtime. Output rises for a few weeks, then fatigue drives up errors and rework, deadlines slip again, and more overtime is approved. Which archetype is this?

Target word: archetype **Decides:** The fix's own side effect (fatigue errors) worsens the original problem.

A	Inverted operation	More of the fix produces more of the problem.
B	Confusion set	No growth engine is hitting a constraint.
C	Correct	A quick fix helps briefly, then its side effect feeds the problem.
D	Adjacent category	That needs a neglected fundamental solution whose capacity erodes; here the fix itself backfires.

8. Answer D Concept 5 · transfer

In the fixes-that-fail archetype, what makes the problem come back?

Target word: reverse: from archetype to mechanism **Decides:** An unintended consequence of the fix, after a delay.

A	Adjacent category	That is shifting the burden.
B	Absolute language	In this archetype the fix does work at first.
C	Confusion set	That is limits to growth.
D	Correct	That delayed side effect is the defining structure.

9. Answer B Concept 6

After a third data breach in two years at a mid-sized insurer ~~with offices in four states~~, leaders learn that staff are rewarded for closing tickets fast, security reviews are optional, and many managers believe “security slows us down.” Using the iceberg model, at which level is “security slows us down”?

Target word: iceberg level of a belief **Decides:** A widely held belief is a mental model.

A	Adjacent category	Rewards and optional reviews are structures; the belief sits beneath them.
B	Correct	Beliefs and assumptions sit at the base of the iceberg.
C	Confusion set	Three breaches in two years is the pattern.
D	Confusion set	The latest breach is the event.

10. Answer A Concept 6 · transfer

Which intervention works at the structure level of the iceberg?

Target word: reverse: from level to intervention **Decides:** Structures are rules, incentives, and processes.

A	Correct	Both change the rules and rewards that produce the pattern.
B	Adjacent category	A response to the event.
C	Confusion set	Watches the pattern but does not change the structure.
D	Inverted operation	Targets mental models, one level deeper than structure.

11. Answer B Concept 7

A hospital ~~with a new parking garage~~ rewards each department for its own efficiency. The ER cuts wait times by moving patients to inpatient units faster, but inpatient units overflow and discharge delays grow; overall patient flow worsens. What principle explains this?

Target word: principle **Decides:** Each part optimized alone; whole got worse.

A	Absolute language	The stem shows the whole got worse.
B	Correct	The ER’s gain became the inpatient units’ overload.
C	Inverted operation	Pushes harder on the local optimization that caused the overflow.
D	Adjacent category	Treats a symptom and ignores the incentive structure.

12. Answer C Concept 8

A meal-kit company’s subscribers grew 15% a month for a year as marketing spend rose. Growth has now stalled even though marketing spending keeps climbing, and delivery complaints have tripled because the single packing center is at capacity. ~~The company’s app has a 4.7 star rating.~~ What should leaders do?

Target word: response to stalled growth **Decides:** Packing capacity is the limiting constraint.

A	Inverted operation	Pushing the reinforcing loop harder only worsens the overload.
B	Absolute language	Growth can resume once the constraint is lifted.
C	Correct	Limits to growth: remove the limit rather than push the engine.
D	Camouflage	The app rating does not address the capacity limit.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Reinforcing vs. balancing loops	Reinforcing loops amplify change (growth or collapse); balancing loops push a system toward a goal.	Calling any loop with a good outcome 'balancing'.	Accelerating growth vs. settling near a target.
2	Delays cause oscillation	A delay inside a balancing loop makes people overcorrect, producing overshoot and swings.	Blaming the swings on outside demand or on people.	Lead times plus a boom-bust pattern.
3	Stocks and flows	A stock changes only by its inflows minus its outflows over time.	Treating the net flow as the stock, or mixing up inflow and outflow.	A starting level plus amounts added and removed.
4	Leverage points	Changing a system's goals, rules, or paradigm has more leverage than adjusting parameters like prices or quantities.	Assuming the biggest physical or numeric change is the strongest lever.	A list of interventions at different depths.
5	Fixes that fail	A quick fix eases a symptom, but its side effect, after a delay, makes the original problem worse.	Confusing it with shifting the burden or limits to growth.	The fix works briefly, then the same problem returns stronger.
6	The iceberg model	Events sit on top of patterns, which come from structures, which rest on mental models.	Treating structures (rules, incentives) as mental models or vice versa.	A belief, a policy, a trend, and a one-time event in the same stem.
7	Emergence and suboptimization	System performance emerges from how parts interact; optimizing each part alone can hurt the whole.	Assuming that if every part improves, the whole improves.	Each unit hits its own target while the overall result gets worse.
8	Limits to growth	When growth stalls against a constraint, relieve the constraint instead of pushing harder on the growth engine.	Spending more on what drove growth before.	Growth flattens while effort rises and a capacity signal appears.

Course scope (public description): Introduction to Systems Thinking and Applications provides learners with the skills required to engage in a holistic systems-based approach to analyzing complex problems and solutions. This course introduces the foundational concepts and principles of systems thinking and provides opportunities to use a systems thinking approach to analyze and evaluate real-world case studies. The course will culminate with using systems thinking to develop a solution to an authentic complex problem. This course has no prerequisites, but general education math (C955 or C957) is preferred. Because the course is self-paced, learners may move through the material as quickly or as slowly as needed, with the goal of demonstrating proficiency in the five competencies covered in the final assessment. If learners have no prior knowledge of this material, they can expect to spend 30 to 40 hours on the course content.